

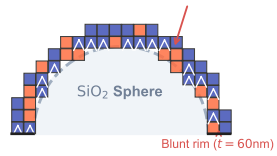
Evolutionary Matrix of Micromagnetic Modeling in FePt Janus Caps

Deconstructing Brown's Paradox: From Coarse-Mesh Discretization Artifacts to Thickness-Dependent Ordering Kinetics & Polycrystalline Scaling

Stage 1: Legacy v1 Model (Thesis / Early Preprint)

Uniform $t = 60\text{nm}$ · Coarse $\Delta x \gg \ell_{ex}$

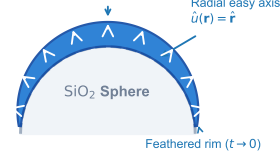
Coarse cells $\Delta x \gg \ell_{ex}$
(Exchange uncoupled)



Stage 2: Continuum Benchmark (Run R1: Single-Crystal Cap)

Ballistic $t_0 \cos \theta$ · Sub-nm Mesh · 100% L_{10}

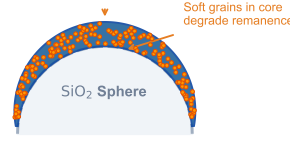
Ballistic taper $t_0 \cos \theta$
Sub-nm mesh $\Delta x = 1.0\text{nm}$



Stage 3: Uniform Disorder (Negative Control M1)

Ballistic $t_0 \cos \theta$ · Sub-nm · Random f_{A1}

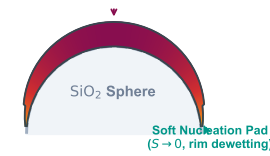
Homogeneous A1 grains
($f_{A1} = 20\text{--}66\%$)



Stage 4: Multiscale Gradient (Kinetic Model M2 / M4)

Kinetic $S(\theta) \propto \sqrt{\cos \theta}$ · Sharrock 300K

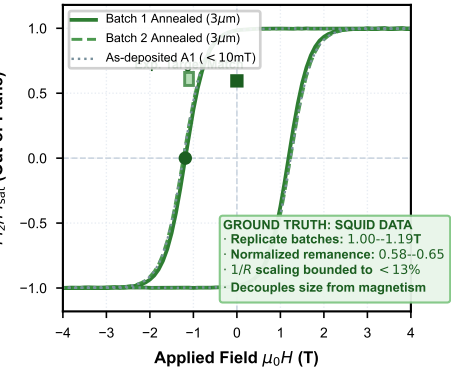
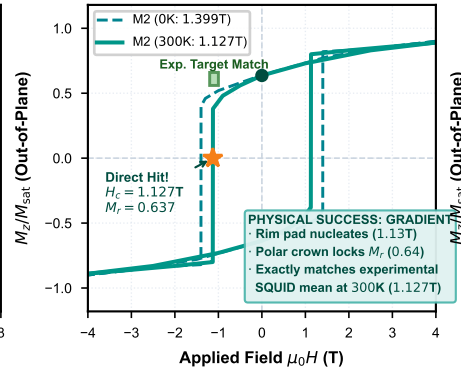
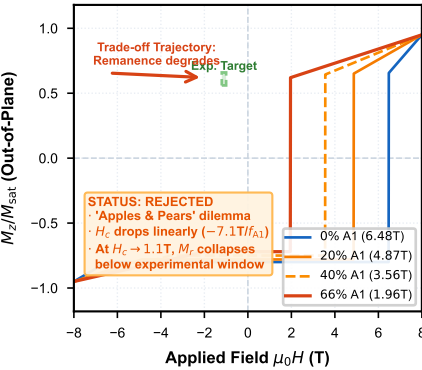
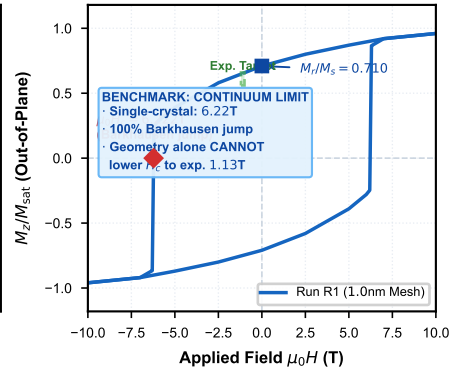
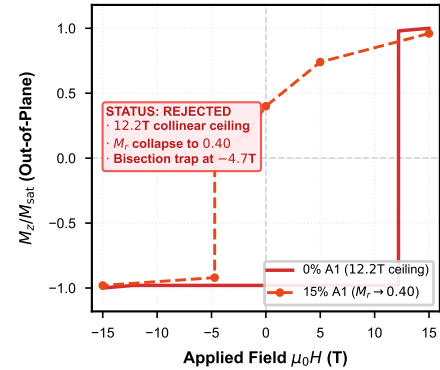
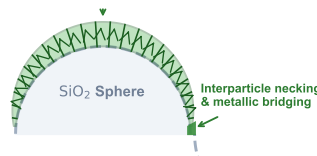
Hard L_{10} Core
($S \geq 0.76$, $K_u \approx 5.3\text{ MJm}^{-3}$)



Stage 5: Experimental Reality (SQUID Ground Truth)

Polycrystalline Caps · 3--10μm · Brown's Paradox

Polycrystalline grains
($d_g \sim 15\text{nm}$, Brown's paradox)



REVERSAL MECHANISM

Uncoupled Cell Flipping
Soft cells reverse independently at low fields;
coarse mesh suppresses domain wall motion.

SIMULATION SPECIFICATION

- [+] Grid Cell Size Δx : 18--92nm ($\gg \ell_{ex}$)
- [+] Thickness Profile: Uniform $t = 60\text{nm}$
- [+] Phase Mixture: Random A1 + L_{10} (0--20%)
- [+] Easy Axis Setup: Uniaxial Collinear $\parallel z$
- [+] Solver Algorithm: Bisection (Trapped)

Primary Limitation:
Discretization artifact: artificial remanence collapse & unphysical 12.2T ceiling

REVERSAL MECHANISM

Uniform Canting & Barkhausen Jump
Moment cants reversibly along radial axes until
a catastrophic single-crystal collapse at 6.22 T.

SIMULATION SPECIFICATION

- [+] Grid Cell Size Δx : 1.0nm ($< \ell_{ex}, \delta_0$)
- [+] Thickness Profile: Ballistic $t_0 \cos \theta$ (60 → 0nm)
- [+] Phase Mixture: 100% Ordered L_{10} ($S = 1.0$)
- [+] Easy Axis Setup: Radial $\hat{u} = \hat{r}$ (85° cut)
- [+] Solver Algorithm: Monotonic Descending Sweep

Primary Limitation:
Single-crystal benchmark: proves geometry alone cannot lower H_c to 1.13 T

REVERSAL MECHANISM

Percolative Soft-Phase Nucleation
Random soft grains trigger uncoordinated reversal;
severe remanence penalty at high A1 fraction.

SIMULATION SPECIFICATION

- [+] Grid Cell Size Δx : 1.3nm ($< \ell_{ex}$)
- [+] Thickness Profile: Ballistic $t_0 \cos \theta$ (60 → 0nm)
- [+] Phase Mixture: Homogeneous $f_{A1} = 10\text{--}66\%$
- [+] Easy Axis Setup: Radial $\hat{u} = \hat{r}$
- [+] Solver Algorithm: Monotonic Descending Sweep

Primary Limitation:
'Apples & Pears' trade-off: high disorder drops H_c but degrades M_r below target

REVERSAL MECHANISM

Coordinated In-situ Rim Nucleation
Thin rim pad initiates reversal; domain wall depins
and sweeps through the hard polar crown.

SIMULATION SPECIFICATION

- [+] Grid Cell Size Δx : 1.3nm ($< \ell_{ex}$)
- [+] Thickness Profile: Ballistic $t_0 \cos \theta$ + dewetted rim
- [+] Phase Mixture: Kinetic $S(\theta) = 0.90\sqrt{\cos \theta}$
- [+] Easy Axis Setup: Radial $\hat{u} = \hat{r}$
- [+] Thermal Correction: Sharrock scaling at 300K

Primary Success:
Direct match: $H_c = 1.127\text{T}$ and $M_r = 0.637$ via thickness kinetics

REVERSAL MECHANISM

Polycrystalline Brown's Paradox
Nucleation at grain boundaries, defect depinning,
and interparticle contact neck pinning.

SIMULATION SPECIFICATION

- [+] Array Dimensions: $D = 3, 5, 8, 10\mu\text{m}$
- [+] Thickness Profile: Line-of-sight $t_0 \cos \theta$
- [+] Film Structure: Polycrystalline ($d_g \sim 15\text{nm}$)
- [+] Easy Axis Setup: Radial template + grain spread
- [+] Boundary Features: Contact necks & interparticle bridging

Primary Success:
Empirical match: monotonic 1/R scaling absent; size decoupled from H_c

3D GEOMETRIC RECONSTRUCTION

Coarse Voxelized Cartesian Mesh



[V] BASELINE PREPRINT (V1)

- [= GEOMETRY] Uniform $t = 60\text{nm}$ (blunt 90° cut)
- [= MESH] Coarse Cartesian 18--92nm ($\gg \ell_{ex}$)
- [= ANISOTROPY] Uniaxial collinear vertical $\parallel z$
- [= ARTIFACT] Suppressed domain-wall physics

3D GEOMETRIC RECONSTRUCTION

Ballistic Cosine Taper + Sub-nm Mesh



[Δ] GEOMETRY + MESH + EASY AXIS

- [+ Δ GEOMETRY] Ballistic taper $t_0 \cos \theta$ (60 → 0nm)
- [+ Δ MESH] Sub-nm mesh $\Delta x = 1.0\text{nm}$ ($< \ell_{ex}$)
- [+ Δ ANISOTROPY] Radial easy axes $\hat{u}(\mathbf{r}) = \hat{r}$
- [= PHASE] 100% L_{10} constant (bound: 6.22T)

3D GEOMETRIC RECONSTRUCTION

Homogeneous A1 Grain Texture



[Δ] PHASE / COMPOSITION (FIXED GEOM)

- [= GEOMETRY] UNCHANGED (taper $t_0 \cos \theta$)
- [= MESH] UNCHANGED (sub-nm $\Delta x = 1.3\text{nm}$)
- [! Δ PHASE] Dispersed soft A1 grains (10--66%)
- [= DILEMMA] Lowering $H_c \rightarrow 1.1\text{T}$ collapses M_r (< 0.30)

3D GEOMETRIC RECONSTRUCTION

Kinetic Gradient $S(\theta)$ + Dewetted Rim

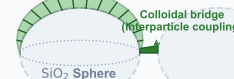


[Δ] GRADIENT S(θ) + THERMAL PHYSICS

- [+ Δ PHASE] Continuous gradient $S(\theta) = 0.90\sqrt{\cos \theta}$
- [+ Δ GEOMETRY] Thermal rim dewetting ($t \rightarrow 0$)
- [+ Δ THERMAL] 300 K Sharrock relaxation (~19.4%)
- [= SUCCESS] Direct match: $H_c = 1.127\text{T}$, $M_r = 0.637$

3D GEOMETRIC RECONSTRUCTION

Polycrystalline Cap & Colloidal Bridge



[Δ] REAL MORPHOLOGY + COLLOIDAL ARRAY

- [= REALITY] Polycrystalline film ($d_g \sim 15\text{nm}$)
- [= COUPLING] Contact necks and interparticle bridges
- [= MECHANISM] Grain-boundary nucleation (Brown real)
- [= SCALING] H_c invariant with curvature radius (<13%)